

PNETLab | Topology

labs.surretnetwork.com/legacy/topology

QEMU (Win2)

```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\User>ncpa.cpl
C:\Documents and Settings\User>ping 50.0.0.10
Pinging 50.0.0.10 with 32 bytes of data:
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Documents and Settings\User>ping 50.0.0.10
Pinging 50.0.0.10 with 32 bytes of data:
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Documents and Settings\User>route print
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ..50 85 5f 09 50 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport
Active Routes:
Network Destination    Netmask          Gateway          Interface        Metric
0.0.0.0                0.0.0.0          60.0.0.1         60.0.0.10        10
60.0.0.10             255.255.255.255  127.0.0.1        127.0.0.1         1
60.255.255.255        255.255.255.255  60.0.0.10        60.0.0.10        10
127.0.0.0             255.0.0.0        127.0.0.1        127.0.0.1         1
224.0.0.0             240.0.0.0        60.0.0.10        60.0.0.10        10
255.255.255.255      255.255.255.255  60.0.0.10        60.0.0.10        1
Default Gateway:      60.0.0.1
Persistent Routes:
None
C:\Documents and Settings\User>
```

Physical

01 : 00 : 00

START

4:23 PM 7/14/2026

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QEMU (Win2)

```
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\User>route print
=====
Interface List
0x2...50 86 5f 09 50 00 ..... MS TCP Loopback interface
0x2...50 86 5f 09 50 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport
=====
Active Routes:
Network Destination        Netmask          Gateway           Interface         Metric
-----
60.0.0.0                   255.0.0.0        60.0.0.0          60.0.0.10         10
60.0.0.10                  255.255.255.255  127.0.0.1         127.0.0.1         10
60.255.255.255            255.255.255.255  60.0.0.10        60.0.0.10         10
127.0.0.0                  255.0.0.0        127.0.0.1         127.0.0.1         1
127.0.0.1                  255.0.0.0        127.0.0.1         127.0.0.1         10
255.255.255.255          255.255.255.255  60.0.0.10        60.0.0.10         1
255.255.255.255          255.255.255.255  60.0.0.10        60.0.0.10         1
=====
Persistent Routes:
None
C:\Documents and Settings\User>ncpa.cpl
C:\Documents and Settings\User>ping 50.0.0.10
Pinging 50.0.0.10 with 32 bytes of data:
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Documents and Settings\User>ping 50.0.0.10
Pinging 50.0.0.10 with 32 bytes of data:
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Documents and Settings\User>
```

Physical

01 : 00 : 00

START

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QEMU (Win1)

```
C:\WINDOWS\system32\cmd.exe

Active Routes:
=====
Network Destination     Netmask          Gateway          Interface        Metric
-----
50.0.0.0                255.0.0.0        50.0.0.10       50.0.0.10       10
50.0.0.10              255.255.255.255  127.0.0.1       127.0.0.1       1
50.255.255.255         255.255.255.255  50.0.0.10       50.0.0.10       10
127.0.0.0              255.0.0.0        127.0.0.1       127.0.0.1       1
224.0.0.0              240.0.0.0        50.0.0.10       50.0.0.10       10
255.255.255.255       255.255.255.255  50.0.0.10       50.0.0.10       1

Persistent Routes:
None

C:\Documents and Settings\User>route delete 60.0.0.0

C:\Documents and Settings\User>route print

Interface List
-----
0x1 ..... MS TCP Loopback interface
0x2...50 fe 9c 09 4f 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:
=====
Network Destination     Netmask          Gateway          Interface        Metric
-----
50.0.0.0                255.0.0.0        50.0.0.10       50.0.0.10       10
50.0.0.10              255.255.255.255  127.0.0.1       127.0.0.1       10
50.255.255.255         255.255.255.255  50.0.0.10       50.0.0.10       10
127.0.0.0              255.0.0.0        127.0.0.1       127.0.0.1       1
224.0.0.0              240.0.0.0        50.0.0.10       50.0.0.10       10
255.255.255.255       255.255.255.255  50.0.0.10       50.0.0.10       1

Persistent Routes:
None

C:\Documents and Settings\User>route print

Interface List
-----
0x1 ..... MS TCP Loopback interface
0x2...50 fe 9c 09 4f 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:
=====
Network Destination     Netmask          Gateway          Interface        Metric
-----
0.0.0.0                 0.0.0.0          50.0.0.1         50.0.0.10       10
50.0.0.0                255.0.0.0        50.0.0.10       50.0.0.10       10
50.0.0.10              255.255.255.255  127.0.0.1       127.0.0.1       10
50.255.255.255         255.255.255.255  50.0.0.10       50.0.0.10       10
127.0.0.0              255.0.0.0        127.0.0.1       127.0.0.1       1
224.0.0.0              240.0.0.0        50.0.0.10       50.0.0.10       10
255.255.255.255       255.255.255.255  50.0.0.10       50.0.0.10       1

Default Gateway:
50.0.0.1

Persistent Routes:
None

C:\Documents and Settings\User>
```

Physical 01 : 00 : 00 START

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New folderGmailYouTubeMaps

QEMU (Win1)

C:\WINDOWS\system32\cmd.exe

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\user>ping 60.0.0.10

Pinging 60.0.0.10 with 32 bytes of data:

Reply from 60.0.0.10: bytes=32 time=1ms TTL=127

Reply from 60.0.0.10: bytes=32 time=1ms TTL=127

Reply from 60.0.0.10: bytes=32 time=1ms TTL=127

Reply from 60.0.0.10: bytes=32 time=2ms TTL=127

Ping statistics for 60.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>route print

Interface List

0x1 MS TCP Loopback interface

0x2 ..50 fe 9c 09 4f 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:

Network DestinationNetmaskGatewayInterfaceMetric

50.0.0.0255.255.255.25550.0.0.1050.0.0.1010

50.0.0.10255.255.255.255127.0.0.1127.0.0.110

50.255.255.255255.255.255.25550.0.0.1050.0.0.1010

127.0.0.0255.255.255.25550.0.0.1050.0.0.101

127.0.0.1255.255.255.25550.0.0.1050.0.0.101

255.255.255.255255.255.255.25550.0.0.1050.0.0.101

Persistent Routes:

None

C:\Documents and Settings\user>route delete 60.0.0.0

C:\Documents and Settings\user>route print

Interface List

0x1 MS TCP Loopback interface

0x2 ..50 fe 9c 09 4f 00 Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

Active Routes:

Network DestinationNetmaskGatewayInterfaceMetric

50.0.0.0255.255.255.25550.0.0.1050.0.0.1010

50.0.0.10255.255.255.255127.0.0.1127.0.0.110

50.255.255.255255.255.255.25550.0.0.1050.0.0.1010

127.0.0.0255.255.255.25550.0.0.1050.0.0.101

127.0.0.1255.255.255.25550.0.0.1050.0.0.101

255.255.255.255255.255.255.25550.0.0.1050.0.0.101

Persistent Routes:

None

C:\Documents and Settings\user>

StartC:\WINDOWS\system...

Physical

01 : 00 : 00

START

Search

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QEMU (Win1)

```
C:\WINDOWS\system32\cmd.exe
Pinging 60.0.0.10 with 32 bytes of data:
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\User>route add 60.0.0.0 mask 255.0.0.0 50.0.0.1
C:\Documents and Settings\User>ping 60.0.0.10
Pinging 60.0.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Documents and Settings\User>ping 60.0.0.10
Pinging 60.0.0.10 with 32 bytes of data:
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=2ms TTL=127
Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\User>route print

Interface List
0x{...} 50 10 36 09 4f 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport

=====
Active Routes:
Network Destination    Netmask          Gateway         Interface        Metric
50.0.0.0                255.0.0.0        50.0.0.10      50.0.0.10        10
50.0.0.10              255.255.255.255  127.0.0.1      127.0.0.1        1
50.255.255.255         255.255.255.255  50.0.0.10      50.0.0.10        10
60.0.0.0                255.0.0.0        50.0.0.1       50.0.0.10        1
127.0.0.0              255.0.0.0        127.0.0.1      127.0.0.1        1
224.0.0.0              240.0.0.0        50.0.0.10      50.0.0.10        10
255.255.255.255       255.255.255.255  50.0.0.10      50.0.0.10        1

=====
Persistent Routes:
None

C:\Documents and Settings\User>
```

Physical

01 : 00 : 00

START

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```
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\User>route print
=====
Interface List
0x1 .....MS TCP Loopback interface
0x2 .....Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport
=====
Active Routes:
Network Destination  Netmask          Gateway          Interface        Metric
50.0.0.0             255.0.0.0         50.0.0.10       50.0.0.10       10
50.0.0.10            255.255.255.255   127.0.0.1       127.0.0.1       1
50.255.255.255       255.255.255.255   50.0.0.10       50.0.0.10       10
127.0.0.0            255.0.0.0         127.0.0.1       127.0.0.1       1
224.0.0.0            240.0.0.0         50.0.0.10       50.0.0.10       10
255.255.255.255     255.255.255.255   50.0.0.10       50.0.0.10       1
=====
Persistent Routes:
None

C:\Documents and Settings\User>show ip route
'show' is not recognized as an internal or external command,
operable program or batch file.

C:\Documents and Settings\User>ping 50.0.0.10

Pinging 50.0.0.10 with 32 bytes of data:

Reply from 50.0.0.10: bytes=32 time<1ms TTL=128
Reply from 50.0.0.10: bytes=32 time<1ms TTL=128
Reply from 50.0.0.10: bytes=32 time<1ms TTL=128
Reply from 50.0.0.10: bytes=32 time<1ms TTL=128

Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Documents and Settings\User>ping 50.0.0.1

Pinging 50.0.0.1 with 32 bytes of data:

Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time<1ms TTL=255
Reply from 50.0.0.1: bytes=32 time<1ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255

Ping statistics for 50.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Documents and Settings\User>ping 50.0.0.2

Pinging 50.0.0.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
```

```
C:\WINDOWS\system32\cmd.exe
Ping statistics for 50.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Documents and Settings\User>ping 50.0.0.2
Pinging 50.0.0.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 50.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\User>
C:\Documents and Settings\User>
C:\Documents and Settings\User>ping 60.0.0.10
Pinging 60.0.0.10 with 32 bytes of data:
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\User>route add 60.0.0.0 mask 255.0.0.0 50.0.0.1
C:\Documents and Settings\User>ping 60.0.0.10
Pinging 60.0.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\User>ping 60.0.0.10
Pinging 60.0.0.10 with 32 bytes of data:
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=1ms TTL=127
Reply from 60.0.0.10: bytes=32 time=2ms TTL=127
Ping statistics for 60.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

```
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ping 60.0.0.1
Pinging 60.0.0.1 with 32 bytes of data:
Reply from 60.0.0.1: bytes=32 time<1ms TTL=255
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time=1ms TTL=255
Reply from 60.0.0.1: bytes=32 time<1ms TTL=255
Ping statistics for 60.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Documents and Settings\user>ping 50.0.0.10
Pinging 50.0.0.10 with 32 bytes of data:
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\Documents and Settings\user>route add 50.0.0.0 mask 255.0.0.0 60.0.0.1
C:\Documents and Settings\user>ping 50.0.0.10
Pinging 50.0.0.10 with 32 bytes of data:
Reply from 50.0.0.10: bytes=32 time=2ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Reply from 50.0.0.10: bytes=32 time=1ms TTL=127
Ping statistics for 50.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>
```



```
C:\WINDOWS\system32\cmd.exe
255.255.255.255 255.255.255.255 50.0.0.10 50.0.0.10 1
=====
Persistent Routes:
None

C:\Documents and Settings\user>route print
=====
Interface List
0x1 ..... MS TCP Loopback interface
0x2 ...50 fe 9c 09 4f 00 ..... Intel(R) PRO/1000 MT Network Connection - Packet Scheduler Miniport
=====
Active Routes:
Network Destination Netmask Gateway Interface Metric
0.0.0.0 0.0.0.0 50.0.0.10 50.0.0.10 10
50.0.0.10 255.255.255.255 127.0.0.1 127.0.0.1 10
50.255.255.255 255.255.255.255 50.0.0.10 50.0.0.10 10
127.0.0.0 255.0.0.0 127.0.0.1 127.0.0.1 1
255.255.255.255 255.255.255.255 50.0.0.10 50.0.0.10 1
=====
Persistent Routes:
None

C:\Documents and Settings\user>show ip route
'show' is not recognized as an internal or external command,
operable program or batch file.

C:\Documents and Settings\user>
```

```
Router
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

50.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    50.0.0.0/8 is directly connected, Ethernet0/0
L    50.0.0.1/32 is directly connected, Ethernet0/0
60.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    60.0.0.0/8 is directly connected, Ethernet0/1
L    60.0.0.1/32 is directly connected, Ethernet0/1
Router#
```